

SCALABILITY & FUTURE PLAN

Date: 18th July 2026	Project Name: Book a Doctor App	Team Lead: Kilari Ushodaya
Domain: Healthcare	Team Members: Shyam Prasad Mantri, Madhu Hasitha Kondeti, Nanda Kumar Madugula, Mantena Sirisha	

Scalability Dimension	Current System State	Proposed Upgrade / Solution
User Load	5,000 Concurrent Users	Kubernetes auto-scaling clusters with Nginx Load Balancer.
Data Storage	SQLite / Single DB	PostgreSQL Read-Replicas with Redis caching layer.
File Storage	Local File Vault	AWS S3 / Azure Blob Storage with automated backups.
Performance	< 300 ms API Latency	CDN static asset caching & WebSockets.